VITA HEALTH

AI-Powered Patient Companion

Concept Note

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1. Executive Summary

Vita Health is an AI-powered patient companion application designed to bridge the gap between clinical consultations and day-to-day patient self-management. It provides patients with a conversational, intelligent interface for understanding their symptoms, adhering to prescribed medication schedules, and interpreting laboratory results in plain language. For healthcare providers, it delivers a real-time dashboard that surfaces at-risk patients, flags missed doses, and reduces the volume of repetitive patient inquiries handled by clinical and administrative staff.

Vita Health is built for deployment through partner clinics, hospitals, and health programs rather than as a standalone consumer product. It operates strictly as a decision-support and adherence tool; it does not diagnose, prescribe, or replace the clinical judgment of a licensed medical professional. Every interaction reinforces this boundary, and all uncertain or high-severity cases are escalated to a human clinician.

Vita Health is developed by Hive XI, an AI automation and creative technology agency. This concept note is prepared to introduce the product idea to potential healthcare partners and to open a conversation around a pilot deployment.

2. Problem Statement

Healthcare delivery in Nigeria operates under significant structural pressures. Patient-to-doctor ratios remain among the highest globally, leaving clinicians with limited time per consultation. Patients frequently leave appointments without fully retaining their dosage instructions, follow-up schedules, or the significance of their test results. This leads to avoidable complications, unplanned readmissions, and a steady stream of repeat phone calls and walk-in queries to clinical and front-desk staff.

Several compounding factors make this worse in the Nigerian context:

- Limited health literacy means many patients cannot correctly interpret lab values or medication labels independently.
- Language barriers persist, as the majority of health information and clinical software is available only in English, leaving Hausa, Yoruba, and Igbo-speaking patients underserved.
- Existing digital tools are either generic international chatbots with no local contextual knowledge, or basic SMS reminder systems with no intelligence, no escalation logic, and no provider-facing visibility.
- Caregiver involvement is common, particularly for elderly patients, but existing tools offer no structured mechanism for caregivers to monitor adherence remotely.

The result is a cycle of poor adherence, preventable deterioration, and increased burden on already-stretched clinical teams.

3. Proposed Solution

Vita Health addresses this problem through three interconnected product layers.

3.1 AI Symptom Guidance

Patients describe symptoms through free text or voice input. The assistant asks structured follow-up questions and categorises the interaction into one of three outcome tiers: self-care information for mild, non-urgent presentations; a recommendation to see a doctor within a defined timeframe for moderate presentations; or an immediate escalation to emergency services for high-severity symptoms. Every response is clearly labelled as AI-generated guidance and not a diagnosis, with transparent reasoning provided in plain, accessible language.

3.2 Medication Adherence and Reminders

Patients or their clinic can input a personalised medication schedule. The app delivers timely reminders and logs whether each dose is taken, skipped, or missed. A visible adherence streak motivates continued engagement. Three or more consecutive missed doses trigger a check-in prompt to the patient, and if unresolved, generate a flag visible to the clinical team on the provider dashboard. Caregivers linked to a patient account receive missed-dose notifications and can view the patient's adherence summary.

3.3 Lab Result Explanation

Patients can input laboratory values and receive a plain-language explanation of what each value generally indicates and whether it falls within a typical reference range. Out-of-range results are visually flagged and paired with a recommendation to discuss findings with a clinician, together with a one-tap option to request a follow-up appointment.

3.4 Clinic-Facing Dashboard

Clinical and administrative staff access a web-based dashboard that shows their enrolled patient panel, with per-patient adherence percentages, last check-in dates, and current status flags. An escalation queue surfaces patients with repeated missed doses or flagged lab results, sorted by urgency, so staff can prioritise outreach efficiently without manually reviewing every patient record.

4. Target Users

The product is designed to serve three distinct user groups simultaneously.

- The patient, who is typically managing a chronic condition such as hypertension, diabetes, or asthma, and requires structured support between clinical visits to maintain their prescribed treatment plan.
- The caregiver, who is often an adult family member managing the health of an elderly or chronically ill relative, frequently from a different location, and who needs visibility into daily medication adherence without constant manual follow-up.
- The clinic administrator or clinician, who oversees a patient panel and needs a simple, real-time tool to identify which patients require follow-up, without it adding significantly to existing documentation or communication workload.

5. AI Safety and Clinical Integrity

Given the sensitivity of health-related AI applications, Vita Health is designed with a robust set of behavioural guardrails that take precedence over all other product considerations.

- The assistant will never present a possible cause or outcome as a confirmed diagnosis.
- All AI-generated guidance is visually distinguished from system messages and carries a persistent disclaimer.
- When a case is ambiguous, complex, or outside the safe scope of the assistant, the system defaults to deferring to a human clinician rather than generating a speculative response.
- A hard-coded emergency detection layer identifies high-severity symptom patterns, such as chest pain combined with shortness of breath or signs of stroke, and immediately redirects the patient to emergency services rather than continuing the conversational flow.
- All triage interactions and outcomes are logged and available to the partner clinic for review and audit.
- The assistant does not alter, suggest changes to, or contradict dosage or treatment instructions set by a clinician.

The deployment model, through a partner clinic rather than direct to consumers, is itself a structural safety choice. Clinical responsibility remains with the partnering institution, and Vita operates under its supervision.

6. Regulatory Positioning

Vita Health is designed to align with the regulatory environment governing healthcare technology in Nigeria. Data collection, storage, and sharing practices are structured in compliance with the Nigeria Data Protection Act (NDPA), including principles of data minimisation, purpose limitation, and explicit patient consent for caregiver-linked data access. Patient health data is encrypted in transit and at rest.

The product is positioned to operate in a supportive, non-diagnostic capacity consistent with guidance from the Medical and Dental Council of Nigeria (MDCN) on the appropriate role of AI-assisted tools in clinical settings. Prior to any deployment, the product will undergo a compliance review with the partnering clinic's own legal and clinical governance team.

7. Technology Approach

Vita Health is a mobile-first application targeting Android devices, which account for the substantial majority of smartphone usage among the target market in Nigeria. The app is designed to perform acceptably on 3G connectivity and handles intermittent network availability gracefully, with medication reminders cached locally and synced when connectivity is restored.

The conversational AI layer is powered by a large language model accessed via API, with a carefully engineered system prompt and structured output schema that enforces the safety and tone guidelines described in Section 5. The symptom triage logic uses a hybrid approach: pattern detection for high-severity emergency keywords operates independently of the language model to ensure reliability, while nuanced symptom exploration is handled conversationally.

The platform supports English and Hausa at launch, with Yoruba and Igbo support planned for Phase 2. Voice input and output are included to lower the barrier for users with limited literacy.

8. Phased Roadmap

Risk	Mitigation
Phase 1 (MVP)	Onboarding and consent flow, AI symptom guidance, medication reminders and adherence tracking, lab result explanation, emergency escalation, caregiver-linked accounts, clinic dashboard, English and Hausa language support. Pilot with one partner clinic.
Phase 2	Appointment booking and calendar integration, Yoruba and Igbo language support, expanded caregiver notification controls, NAFDAC drug registration lookup, USSD and offline reminder mode.
Phase 3	EMR/EHR integration for partner hospitals, wearable device vitals integration, NHIS and health insurance linkage, mental health check-in module.

9. Success Metrics

The following primary metrics will be used to evaluate a pilot deployment over a 90-day period.

- Change in medication adherence rate among enrolled patients, measured against baseline data from the partner clinic.

- Reduction in adherence-related phone calls and unplanned walk-in visits to the partner clinic.
- Percentage of flagged alerts on the clinic dashboard reviewed by clinical staff within 48 hours.
- Patient-reported comprehension score from a short in-app survey measuring whether users understood their recommended next steps after each key interaction.
- Caregiver satisfaction with visibility and notification quality, measured via a post-pilot interview or survey.

10. Risks and Mitigations

Risk	Mitigation
Patient over-reliance on AI guidance	Persistent on-screen disclaimers, conservative escalation thresholds, and language review of all AI prompts to ensure caution is communicated clearly.
Missed emergency presentation	Hard-coded emergency pattern detection that bypasses the language model and triggers immediately, independently of the conversational flow.
Low adoption due to data cost or device limitations	Lightweight app build optimised for low-bandwidth networks, locally cached reminders for offline functionality, and a phased plan for USSD support.
Clinic staff distrust of AI-generated content	Full audit log of all AI interactions visible to clinical staff, transparent reasoning displayed alongside every AI suggestion, no autonomous action taken without human review.
Patient data privacy concerns	Explicit consent flow at onboarding, opt-in caregiver sharing, NDPA-aligned data handling, and encryption at rest and in transit.

11. About Hive XI

Hive XI is a remote-first AI automation and creative technology agency. The agency builds AI-powered products, workflow automations, and digital systems for clients across healthcare, education, logistics, and enterprise sectors. Vita Health represents Hive XI's flagship demonstrator in the health technology vertical, developed to address a clearly identified gap in patient-facing digital infrastructure in Nigeria.

Hive XI is interested in partnering with forward-looking clinics, hospitals, and health programs who share the vision of extending quality healthcare support beyond the clinic walls and into the daily lives of patients.

12. Next Steps

This concept note is intended to open a dialogue. Hive XI invites interested healthcare partners to engage in a discovery conversation covering the following questions.

- Which patient populations at your institution would benefit most from an adherence and symptom-guidance companion?
- What existing systems, if any, would an MVP integration need to connect with?
- What are the institution's data governance and patient consent requirements for a digital health tool of this kind?
- What would a realistic pilot scope and timeline look like within your operational context?

To schedule a conversation, please contact Hive XI directly. A live walkthrough of the product wireframes and the full Product Requirements Document are available on request.